

MongoDB and Vercel Deployment Guide

A step-by-step roadmap for moving the Student Progress & Evaluation Platform from local SQLite to MongoDB Atlas and deploying it on Vercel.

Prepared for the admin-only application baseline.

Key decision: Prisma 7 does not support MongoDB. This guide uses Prisma 6 for MongoDB. PostgreSQL remains the lower-risk option for this relational application.

1. Roadmap and Architecture Decision

Complete the migration in this order:

- Create a MongoDB Atlas cluster and database user.
- Downgrade Prisma from version 7 to the latest Prisma 6 release.
- Convert the schema provider from SQLite to MongoDB.
- Convert IDs and relation fields to MongoDB ObjectIds.
- Replace the SQLite Prisma adapter with the classic Prisma client.
- Use `prisma db push` instead of SQL migrations.
- Seed only the admin account.
- Add production environment variables in Vercel.
- Deploy, test login, and verify database persistence.

MongoDB is document-oriented, but this application has many relational models: users, batches, submissions, tests, feedback, evidence, reports, and audit logs. Expect a schema conversion, not a one-line provider change.

2. Create MongoDB Atlas Database

- Open mongodb.com/atlas and create a free cluster.
- Create a database user with a strong password.
- Under Network Access, allow Vercel connections. For a first deployment, `0.0.0.0/0` works, but restrict access where possible.
- Choose Connect, Drivers, and copy the MongoDB connection string.

```
mongodb+srv://edtech_admin:PASSWORD@cluster.mongodb.net/edtech?retryWrites=true&w=majority
```

3. Install Prisma MongoDB Dependencies

```
cd C:\Users\thakk\Downloads\webbbb
npm uninstall @prisma/client prisma @prisma/adapters-better-sqlite3 better-sqlite3
npm install @prisma/client@6 prisma@6
npm install -D tsx
```

MongoDB projects should remain on Prisma 6. Do not use Prisma 7 SQL driver adapters for MongoDB.

4. Update Prisma Schema Header

In `prisma/schema.prisma`, use:

```
generator client {
  provider = "prisma-client-js"
}

datasource db {
  provider = "mongodb"
  url      = env("DATABASE_URL")
}
```

5. Convert IDs to ObjectIds

MongoDB model IDs should use:

```
id String @id @default(auto()) @map("_id") @db.ObjectId
```

Every relation scalar must use `@db.ObjectId` too:

```
userId String @db.ObjectId
batchId String @db.ObjectId
someId String? @db.ObjectId
```

Apply this to User, Batch, UserBatch, Announcement, Submission, SubmissionFile, Test, Question, TestResponse, Answer, Evidence, Feedback, KPI, Report, and AuditLog.

6. Replace the Prisma Client Setup

Replace the SQLite adapter in `src/lib/prisma.ts` with:

```
import { PrismaClient } from "@prisma/client";

const globalForPrisma = globalThis as unknown as {
  prisma: PrismaClient | undefined;
};

export const prisma = globalForPrisma.prisma ?? new PrismaClient({
  log: process.env.NODE_ENV === "development"
    ? ["error", "warn"]
    : ["error"],
});

if (process.env.NODE_ENV !== "production") {
  globalForPrisma.prisma = prisma;
}
```

7. Configure Local Environment

```
DATABASE_URL="mongodb+srv://edtech_admin:YOUR_PASSWORD@cluster.mongodb.net/edtech?
retryWrites=true&w=majority"
AUTH_SECRET="replace-with-a-long-random-secret"
```

Do not commit `.env` or database credentials.

8. Generate, Push, and Seed

```
npx prisma generate
npx prisma db push
npx tsx prisma/seed.ts
npx prisma studio
```

MongoDB does not use the existing SQL migration workflow. Use `prisma db push`. The current seed script resets demo records and creates only `admin@sou.edu`.

9. Configure Vercel

In Vercel, open Project Settings, Environment Variables, and add these for Production, Preview, and Development:

```
DATABASE_URL=mongodb+srv://edtech_admin:YOUR_PASSWORD@cluster.mongodb.net/edtech?
retryWrites=true&w=majority
AUTH_SECRET=your-production-secret
```

Redeploy after saving:

```
git add -A
git commit -m "Migrate database from SQLite to MongoDB"
git push origin main
```

10. Test the Deployment

- ☐ Open /login.
- ☐ Sign in with admin@sou.edu.
- ☐ Confirm the redirect to /admin.
- ☐ Create a real faculty or student from the admin panel.
- ☐ Refresh and confirm the record persists.
- ☐ Check Vercel Function Logs if login returns HTTP 500.

11. File Storage Requirement

MongoDB only handles database records. Vercel local disk is not persistent for uploads and generated PDFs. Attach Vercel Blob and set:

```
BLOB_READ_WRITE_TOKEN=your-vercel-blob-token
```

12. Troubleshooting

- HTTP 500 on login: check DATABASE_URL and AUTH_SECRET in Vercel.
- Prisma client error: run `npm install` and `npx prisma generate`.
- Connection timeout: check MongoDB Atlas Network Access.
- Authentication fails: run the admin-only seed against the same production DATABASE_URL.
- Data disappears: confirm the app is using MongoDB, not a local SQLite fallback.